

Khush A. Patel

Harrisburg, Pennsylvania

kpate151@jhu.edu — U.S. Citizen
www.linkedin.com/in/khushpatelengineer/

Objective

Johns Hopkins Mechanical Engineering student seeking to apply and learn skills in electromechanical design and simulation.

Education

Johns Hopkins University, Whiting School of Engineering

Baltimore, MD

B.S. Mechanical Engineering **GPA: 4.00/4.00**

Anticipated May 2029

Relevant Coursework: Mechanics, Multivariable Calculus, Differential Equations, Electricity & Magnetism, Digital Electronics, CAD (SolidWorks, Creo), Python.

Technical Skills

- **CAD & FEA + CFD:** SolidWorks CAD and Static FEA, Creo Parametric, ANSYS Mechanical, Fluent, CFX, OpenFOAM
- **Programming and Data:** Python (NumPy, SciPy, Matplotlib), R, Basic C++,
- **Robotics & Controls:** ROS 2, kinematics, torque analysis, PID, LQR, MPC; motion planning and trajectory control
- **Manufacturing:** Dimensioned Drawings, Basic GD&T, DFM/DFA (interference analysis), 3D Printing, Laser Cutting
- **Electronics:** MultiSim, PLDs, Microcontrollers, Encoders, Soldering, FFT analysis, Basic systems theory (BIBO stability, SISO)

Engineering Experience

Johns Hopkins University – AIRO Laboratory

Baltimore, MD

Undergraduate Researcher – Controls & CFD

Dec 2025–Present

- Leading **CFD wake modeling** of a naval vessel, owning simulation setup, execution, and validation.
- Optimized **OpenFOAM** workflows for accurate wake data (turbulence modeling, mesh quality, refinement, y+).
- Integrated CFD data into Python ROS2 pipelines for stochastic MPC to improve drone landings on moving ships.

Amarkosha Technologies (Start-Up)

Baltimore, MD

Mechanical Lead – Product Development

Jul 2025–Dec 2025

- Led mechanical design optimization of a vibration-induced energy harvester.
- Designed initial coil–magnet system for mechanical-to-electrical energy conversion.
- Performed transient CFD and FSI (**ANSYS Fluent/CFX + Mechanical**) to predict aeroelastic response under different materials.
- Applied FFT analysis to lift-force data to characterize vortex shedding; selected for SPARK Accelerator + BII Grant finalist.

Johns Hopkins Mars Rover Team

Baltimore, MD

Mechanical & Controls Engineer – Manipulator

Sep 2025–Present

- Designed a compact **3-DOF robotic wrist**; performed joint torque calculations to select optimal BLDC motors under budget constraint.
- Designed, tested, and iterated a motor mount, housing, and encoder integration to a **cycloidal gearbox** design.

GreenWorks Development

Mechanicsburg, PA

Project Development Intern

Jan–Aug 2025

- Successfully analyzed commercial solar systems' savings and cost models.
- Pitched multi-million-dollar proposals to two major corporate clients.

Projects

4-DOF Robotic Arm (Independent)

- Designed and built a 4-DOF robotic arm with CAD-based design and mechanical assembly.
- Simulated motion in Webots with inverse kinematic control; integrated Arduino actuation and encoder feedback.
- Applied static FEA and DFM principles for structural reliability.

Leadership & Honors

- **Founder & Co-President, Renewable Energy Club (RENEW); Facilities Chair, Mini-THON** — \$280K pediatric cancer fundraiser
- **Full-Ride Institutional Scholarship (JHU);** William R. Pierce Physics Award; Melissa Huang Scholarship; CVHS Science Hall of Fame